Coding & Solution

Date	28 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[GitHub]
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Bootstrap 5
Key Features Implemented	• Crop recommendation using Machine Learning • Data preprocessing and feature selection • Random Forest model training and prediction • Responsive Flask web application • Input validation and error handling • Model persistence using Pickle (.pkl)
Pending / Incomplete Features	• Fertilizer recommendation module • Weather API integration • User authentication and login • Cloud deployment (AWS/Azure) • Mobile application support
Setup / Run Instructions	1. Clone the repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Run the application using python app.py. 5. Open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Comments:

The OptiCrop application follows a modular architecture with separate components for data preprocessing, machine learning model training, Flask backend development, and frontend presentation. The solution is designed to be maintainable and scalable, allowing future enhancements such as fertilizer recommendation, weather forecasting, IoT integration, and cloud deployment.